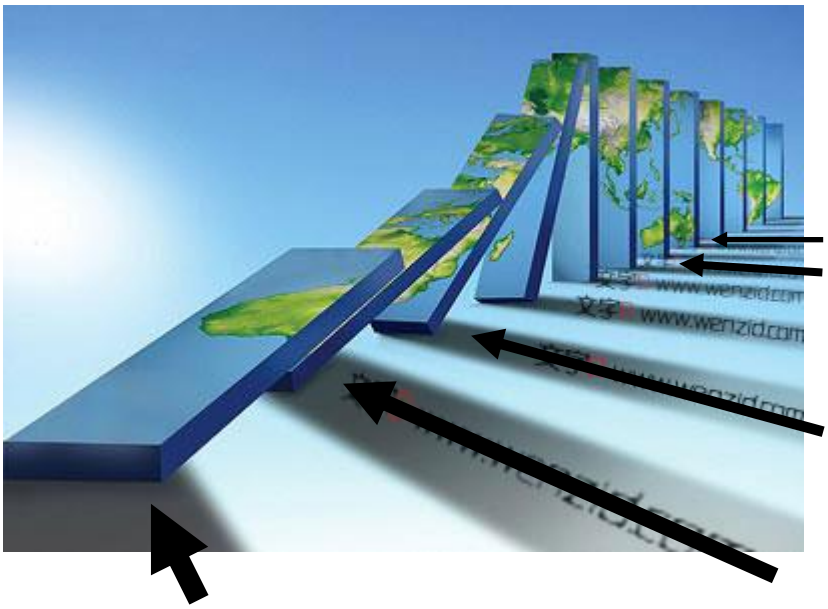
Mathematical Induction (數學歸納法)

The principal of Mathematical Induction is similar to domino (骨牌):



(k+1)th domino: P(k+1)

k^{th} domino: P(k)

3rd domino: P(3)

1st domino: denoted as P(1)

2nd domino: P(2)

How to make sure that all the dominoes will fall? The following conditions must be satisfied:

1. The first domino **P(1)** falls
2. If any one domino **P(k)** falls, the domino next to it **P(k+1)** will also fall.

To apply the above principle to prove propositions (命題) related to positive integers is called Mathematical Induction **(數學歸納法).**

4 Major steps in Mathematical Induction:

For any proposition **P(n)** where n is any positive integer, **P(n)** will be true if

Step 1: **P(1) is true**； (when n = 1, L.H.S. = R.H.S.)

Step 2: **Assume P(k) is true, where k is a positive integer.**

Step 3: **P(k+1) is also true**； (when $n = k + 1$, L.H.S. = R.H.S.)

Step 4: Conclusion. (P(n) is true for all positive integers n.)

Not for sale



Mathematical Induction

數學歸納法：基本類型題目（一）

e.g. Proof , by mathematical induction,

$$1+2+3+...+n=\frac{n(n+1)}{2}$$ for all positive integers n.

Let P(n) be the proposition (命題)

"$1+2+3+...+n=\frac{n(n+1)}{2}$" for all positive integers n.

Step 1: When $n = 1$, L.H.S. = 1

R.H.S. $=\frac{1(1+1)}{2}=1$

$\therefore$ L.H.S. = R.H.S.

P(1) is true .

Step 2: Assume P(k) is true, where k is positive integer.

i.e. $1+2+3+...+k=\frac{k(k+1)}{2}$

Step 3: When $n = k + 1$,

L.H.S. $= 1+2+3+...+k+(k+1)$

$=\frac{k(k+1)}{2}+(k+1)$

$=\frac{k(k+1)}{2}+\frac{2(k+1)}{2}$

$=\frac{(k+2)(k+1)}{2}$

$=\frac{(k+1)[(k+1)+1]}{2}$ = R.H.S.

不要胡亂拆括號，只需抽 Common factor 及通分母即可

$\therefore$ P(k + 1) is also true.

Step 4: By M.I., P(n) is true for all positive integers n.

Not for sale

Mathematical Induction

[A.Math 1989 Paper II Q.2]

Prove, by mathematical induction,

$1\times2\times3 + 2\times3\times4 + \ldots + n(n+1)(n+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$

for all positive integers n. ***(5 marks)***

Not for sale

Mathematical Induction

數學歸納法：基本類型題目（二）

e.g. Proof , by mathematical induction,

$$\frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^3}+\cdots+\frac{n}{3^n}=\frac{1}{4}\left(3-\frac{2n+3}{3^n}\right)$$ for all positive integers n.

Let P(n) be

"$\frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^3}+\cdots+\frac{n}{3^n}=\frac{1}{4}\left(3-\frac{2n+3}{3^n}\right)$" for all positive integers n.

Step 1: When $n = 1$, R.H.S. $= \frac{1}{4}\left(3-\frac{2+3}{3}\right) = \frac{1}{3}$ = L.H.S.

$\therefore$ P(1) is true .

Step 2: Assume P(k) is true, where k is positive integer.

i.e. $\frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^3}+\cdots+\frac{k}{3^k}=\frac{1}{4}\left(3-\frac{2k+3}{3^k}\right)$

Step 3: When $n = k + 1$,

$$\text{L.H.S.} = \frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^3}+\cdots+\frac{k}{3^k}+\frac{k+1}{3^{k+1}}$$

$$= \frac{1}{4}\left(3-\frac{2k+3}{3^k}\right)+\frac{k+1}{3^{k+1}}$$

$$= \frac{1}{4}\left(\frac{3\cdot 3^{k+1}}{3^{k+1}}-\frac{3(2k+3)}{3^{k+1}}\right)+\frac{1}{4}\cdot\frac{4(k+1)}{3^{k+1}}$$

$$= \frac{3^{k+2}-3(2k+3)+4(k+1)}{4(3^{k+1})}$$

$$= \frac{3^{k+2}-2k-5}{4(3^{k+1})}$$

$$= \frac{1}{4}\left[3-\frac{2(k+1)+3}{3^{k+1}}\right] = \text{R.H.S.}$$

不要胡亂拆括號，首先需通分母，然後化簡便可

$\therefore$ P(k + 1) is also true.

Step 4: By M.I., P(n) is true for all positive integers n.

Not for sale

Mathematical Induction

[A.Math 2003 Q.7]

Prove, by mathematical induction that

$\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\cdots+\frac{n}{2^n}=2-\frac{n+2}{2^n}$ for all positive integers n. ***(5 marks)***

Not for sale



Mathematical Induction (Tricky Type)

數學歸納法：進階題型

- **第一項 (First term) 有古怪題型！**

當 $n = 1$ 時，左方不是等於 1，而是等於 $1 + (x + 1)$！

e.g. If $x \neq 0$, proof that

$$1+(x+1)+(x+1)^2+\cdots+(x+1)^n = \frac{(x+1)^{n+1}-1}{x}$$ for all positive integers n.

Let P(n) be

"$1+(x+1)+(x+1)^2+\cdots+(x+1)^n = \frac{(x+1)^{n+1}-1}{x}$" for all positive integers n.

Step 1: When $n = 1$,

L.H.S. $= 1 + (x + 1) = x + 2$

R.H.S. $= \frac{(x+1)^{1+1}-1}{x}$

$= \frac{x^2+2x+1-1}{x} = x + 2$

$\therefore$ P(1) is true.

Step 2: Assume P(k) is true, where k is positive integer.

i.e. $1+(x+1)+(x+1)^2+\cdots+(x+1)^k = \frac{(x+1)^{k+1}-1}{x}$

Step 3: When $n = k + 1$,

L.H.S.

$= 1+(x+1)+(x+1)^2+\cdots+(x+1)^k + (x+1)^{k+1}$

$= \frac{(x+1)^{k+1}-1}{x} + (x+1)^{k+1}$

$= \frac{(x+1)^{k+1}-1}{x} + \frac{x(x+1)^{k+1}}{x}$

$= \frac{(x+1)(x+1)^{k+1}-1}{x}$

$= \frac{(x+1)^{[(k+1)+1]}-1}{x}$ = R.H.S.

不要胡亂拆括號，只需通分母及化簡即可

$\therefore$ P($k + 1$) is also true.

Step 4: By M.I., P(n) is true for all positive integers n.

Not for sale

Mathematical Induction (Tricky Type)

[A.Math 2007 Q.5]

Let $a \neq 0$ and $a \neq 1$. Prove, by mathematical induction that

$\frac{1}{a-1} - \frac{1}{a} - \frac{1}{a^2} - \cdots - \frac{1}{a^n} = \frac{1}{a^n(a-1)}$ for all positive integers n. ***(5 marks)***

Not for sale



Mathematical Induction

數學歸納法： 常見錯誤！！！

☒ Using notation such as P(n), P(k), S(n) and S(k) without defining them first;
在未經定義下，採用 P(n), P(k), S(n) 及 S(k) 等符號；

☒ Making improper statements such as:
給出不準確的句子如：

- "Let n = 1 be true" 「設 n = 1 為真」
- "Assume n = k is true" 「假設 n = k 為真」
- "Assume P(k) is true for any k" 「假設對任意 k，P(k)為真」

☒ Stating conclusion such as "the proposition is true for all integers k."
誤作結論如「命題對所有整數 k 為真」

考考你：下列證明中哪裏有錯？

e.g. Proof by M.I. that $1+2+3+...+n=\frac{n(n+1)}{2}$ for all positive integers n .

When n = 1, L.H.S. = 1

$$\text{R.H.S.} = \frac{1(1+1)}{2} = 1$$

$\therefore$ P(1) is true .

When $n = k + 1$,

$$\text{L.H.S.} = 1+2+3+...+k+(k+1)$$

$$= \frac{k(k+1)}{2}+(k+1)$$

$$= \frac{(k+2)(k+1)}{2}$$

$$= \frac{(k+1)[(k+1)+1]}{2} = \text{R.H.S.}$$

$\therefore$ P(k + 1) is also true.

By M.I., P(k) is true for all integers k.

HKDSE Questions (level 1)

1. ***[M2 2012 Q.3]***

Prove, by mathematical induction, that for all positive integers n,

$1\times 2+2\times 5+3\times 8+\cdots n(3n-1)=n^2(n+1)$. ***(5 marks)***

Not for sale

HKCEE Questions (level 1)

2. ***[A.Math 1993 Paper II Q.1]***

Prove, by mathematical induction that

$$1^2\times 2 + 2^2\times 3 + \ldots + n^2(n+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$$

for any positive integers n. ***(5 marks)***

Not for sale

HKCEE Questions (level 1)

3. ***[A.Math 2000 Paper II Q.4]***

Prove, by mathematical induction,

$1^2 - 2^2 + 3^2 - 4^2 + \ldots + (-1)^{n-1}n^2 = (-1)^{n-1}\dfrac{n(n+1)}{2}$ for all positive integers n.

(6 marks)

Not for sale

HKCEE Questions (level 2)

4. ***[A.Math 1994 Paper II Q.5]***

Prove, by mathematical induction that

$\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}+...+\frac{2n-1}{2^n}=3-\frac{2n+3}{2^n}$ for any positive integers *n*. ***(5 marks)***

Not for sale

HKCEE Questions (level 2)

5. ***[M2 2013 Q.3]***

Prove, by mathematics induction, that for all positive integers n,

$$1+\frac{1}{1\times 4}+\frac{1}{4\times 7}+\frac{1}{7\times 10}+...+\frac{1}{(3n-2)\times(3n+1)}=\frac{4n+1}{3n+1}$$ ***(5 marks)***

Not for sale

HKCEE Questions (level 2)

6. ***[A.Math 2005 Q.8]***

Prove, by mathematical induction that

$\frac{1\times 2}{2\times 3}+\frac{2\times 2^2}{3\times 4}+\frac{3\times 2^3}{4\times 5}+\cdots+\frac{n\times 2^n}{(n+1)(n+2)}=\frac{2^{n+1}}{n+2}-1$ for all positive integers n.

(5 marks)

Not for sale

HKCEE Questions (level 3)

7. ***[A.Math 1990 Paper II Q.2]***

Let $T_n = n^2 + n$ for any positive integer n.

Prove, by mathematical induction,

$T_1+T_2+\ldots+T_n = \frac{1}{3}n(n+1)(n+2)$ for any positive integer n. ***(5 marks)***

Not for sale

HKCEE Questions (Level 3)

8. ***[A.Math 1997 Paper II Q.7]***

Let $T_n = (n^2 + 1)(n!)$ for any positive integers n. Prove, by mathematical induction that $T_1 + T_2 + \ldots + T_n = n[(n+1)!]$ for all positive integers n.

[Note: $n! = n(n-1)(n-2)\ldots 3 \cdot 2 \cdot 1$.] ***(6 marks)***

Not for sale

HKCEE Questions (level 3)

9. ***[A.Math 2001 Q.12]***

Prove, by mathematical induction,

$1\times2 + 2\times3 + 3\times4 + \ldots + n(n+1) = \frac{1}{3}n(n+1)(n+2)$ for all positive integers n.

Hence, evaluate $1\times3 + 2\times4 + 3\times5 + \ldots + 50\times52$. ***(8 marks)***

Not for sale

HKCEE Questions (level 3)

10. ***[A.Math 2002 Q.12]***

(a) Prove, by mathematical induction that for all positive integers n.

$$2(2) + 3(2^2) + 4(2^3) + \cdots + (n + 1)(2^n) = n(2^{n+1})$$

(b) Show that $1(2) + 2(2^2) + 3(2^3) + \cdots + 98(2^{98}) = 97(2^{99}) + 2$.

(8 marks)

Not for sale

HKDSE Questions (level 4)

11. ***[M2 2016 Q.5]***

(a) Using mathematics induction, prove that $\sum_{k=1}^{n}(-1)^k k^2 = \frac{(-1)^n n(n+1)}{2}$

for all positive integers *n*.

(b) Using (a), evaluate $\sum_{k=3}^{333}(-1)^{k+1}k^2$. ***(6 marks)***

Not for sale

HKDSE Questions (level 4)

12. ***[M2 2018 Q.6]***

(a) Using mathematics induction, prove that $\sum_{k=1}^{n} k(k+4) = \frac{n(n+1)(2n+13)}{6}$

for all positive integers *n*.

(b) Using (a), evaluate $\sum_{k=333}^{555} (\frac{k}{112})(\frac{k+4}{223})$. ***(7 marks)***

Not for sale

HKDSE Questions (level 4)

13. ***[M2 2019 Q.5]***

(a) Using mathematics induction, prove that $\sum_{k=n}^{2n} \frac{1}{k(k+1)} = \frac{n+1}{n(2n+1)}$

for all positive integers n.

(b) Using (a), evaluate $\sum_{k=50}^{200} \frac{1}{k(k+1)}$. ***(7 marks)***

Not for sale

HKDSE Questions (level 4)

14. ***[M2 2020 Q.5]***

(a) Using mathematics induction, prove that $\sum_{k=1}^{n} \frac{1}{k(k+1)(k+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$

for all positive integers n.

(b) Using (a), evaluate $\sum_{k=4}^{123} \frac{50}{k(k+1)(k+2)}$. ***(7 marks)***

Not for sale



More Questions in Past Papers

1. ***[A.Math 1988 Paper II Q.5]***

Prove, by mathematical induction,

$1^2 + 3^2 + \ldots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$ for all positive integers n.

(6 marks)

2. ***[A.Math 1992 Paper II Q.1]***

Prove, by mathematical induction that

$1\times2 + 2\times5 + 3\times8 + \ldots + n(3n-1) = n^2(n+1)$ for all positive integers n.

(5 marks)

3. ***[A.Math 1998 Paper II Q.3]***

Prove, by mathematical induction,

$1\times2 + 2\times3 + 2^2\times4 + \ldots + 2^{n-1}(n+1) = 2^n(n)$ for all positive integers n.

(5 marks)

4. ***[A.Math 1986 Paper II Q.1]***

Prove, by mathematical induction that

$\frac{1}{1\times2} + \frac{1}{2\times3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for all positive integers n.

(5 marks)

5. ***[A.Math 1985 Paper II Q.2]***

Let $T_n = \frac{n(n+2)}{(n+1)^2}$ for any positive integers n.

Prove, by mathematical induction that

$$T_1 \times T_2 \times \ldots \times T_n = \frac{n+2}{2(n+1)} \quad \text{for all positive integers } n.$$

(5 marks)

6. ***[A.Math 1991 Paper II Q.7]***

(a) Prove by mathematical induction,

$1^2 + 2^2 + \ldots + n^2 = \frac{1}{6}n(n+1)(2n+1)$ for all positive integers n.

(b) Using the formula in (a), find the sum $1\times2 + 2\times3 + \ldots + n(n+1)$.

(8 marks)

Title of the publication:	Mathematical Induction
Copyright owner:	HKEAA
Date of making this copy:	1 – 9 – 2023
Some material in this note has been copied under a license issued by the HKEAA. You are not permitted to make any further copy of this work, or to make It available to others. It is important to understand and respect copyright.	